

Integers, Place Value and Arithmetic

Module 2 · Teacher Lesson Plan · Gauss Contest Preparation

Original Concept Mastery items. Not reproduced from CEMC or any third party.

1. Module header

Field	Value
Module	2 — Integers, Place Value and Efficient Arithmetic
Hours · week	4.0 h (two 90-minute lessons + one 60-minute extension) · Week 2
Teaching mode	A — Secure: write and teach for speed and zero error
Share of contest points	7.6% of observed Gauss G7 points (2021–2026 analysis)
Curriculum	Ontario Mathematics (2020), Grade 7 — Strand B Number: B1 Number Sense, B2 Operations (expectation-level codes: verify against the official document; the CM code map has no verified Grade 7 Number rows yet).
Canonical topics	Whole Numbers & Place Value · Integers · Decimals · Multiplication & Division · Time & Money
Past-paper reference	Structures only, never printed: https://cemc.uwaterloo.ca/contests/gauss
Student goal	Answer integer and place-value questions fast, with zero errors.
Files	Sheet (student/teacher) · Homework + answer strip + solutions · Drill Book: Integer and Place-Value Fluency

2. Why this module carries this weight

This category is worth 7.6% of observed Gauss Grade 7 points, and 12 of its 13 observed questions sit in Part A (5 points each). That makes it a speed-and-accuracy module, not a depth module: the 4.0 hours buy volume, not difficulty. Teach for zero errors at 30 seconds per item; the single Part C structure here ($10^n - k$) gets one focused block in Lesson B.

3. Prerequisites

Earlier modules. M1 (six core moves, error codes, scoring geometry). Grade 6: integers on a number line, decimal place value to thousandths, whole-number operations.

Five-minute check. First five minutes: (1) Order -3 , 2 , -7 , 0 from least to greatest. (2) Which is larger, 0.4 or 0.39 ? (3) $48 + 25 = ?$ If any item is missed, run the struggling path for that item type.

4. Lesson breakdown

Lesson A (90 min)

Min	Phase	What happens
0-10	Retrieval	Prerequisite check (below) + 3 quick M1 items: What are Part A, B and C worth? Name the six core moves. What error code is 'I misread the question'?
10-25	Concept / strategy	Worked Example 1 (restructure before you calculate). Script below. Introduce the 10-second scan.
25-45	Guided practice	Sheet Q1-6. Student says the scaffold step aloud before writing.
45-70	Paired → independent	Sheet Q7-10 in pairs (12 min, 30 s each target then discuss), then Drill Book Set 1 alone (8 min, target 10 min for 20 items — stop at 8 and note how far).
70-80	Method comparison	Worked Example 2: calculate-each-sum vs pair-the-terms. Class votes, then times both.
80-90	Exit check (Lesson A)	Two items on a slip: (1) $25 \times 37 \times 4 = ?$ (2) Which is farther from 0, -6.5 or 6.05 ? Record accuracy; anything below 2/2 goes into Lesson B retrieval.

Lesson B (90 min)

Min	Phase	What happens
0-10	Retrieval	Redo the Lesson A exit items with new numbers: $125 \times 29 \times 8$; farther from 0: -3.7 or 3.69 . Two Drill Set 1 items the student missed.
10-25	Concept / strategy	$10^n - k$ and time/money regrouping. Build the $10^n - 47$ table as a class; model $7:48 + 2:37$ by making the next hour.
25-45	Guided practice	Drill Book Set 2 timed (10 min target), then mark and code errors together.
45-70	Paired → independent	Sheet Q11-14 independent (8 min, 30 s each), then Q15-16 in pairs (12 min) using the hint ladder. Record hint level used.
70-80	Method comparison	Q15: small-case table vs ' $10^{30} - 47 = (10^{30} - 1) - 46$ '. Which transfers to Q16 and HW Q10?
80-90	Exit ticket	Sheet Q17-18. Hand out homework; set the 25-minute time target.

Extension session (60 min)

Min	Phase	What happens
0-10	Retrieval	Five errors from the student's error log, re-worked cold.
10-30	Timed accuracy	Drill Book Sets 3 and 4 back to back. Target: 90% at 30 s per item.
30-50	Error analysis	Code every miss (C R S A V Q T K). For each A or Q error, write the check that would have caught it.
50-60	Speed round	Mixed Final Set items 1-10 against the clock; compare with Set 1 time.

5. Teaching scripts for the worked examples

Scripts are in second person to you; student lines are in quotes. Pause means wait — silence is the tool.

Worked Example 1 — Restructure before you calculate

- **Board:** write only the problem. **Say:** “Pencils down. You have ten seconds to look, not to calculate. What do you notice?” **Pause** the full ten seconds.
- **Listen for three levels.** Not yet: “36 times 7 is 252...” → “Hold that. What is the same in both products?” Developing: “They both have a 7.” → “What could you do with that 7?” Secure: “36 and 64 make 100, so it’s 700.” → “Prove it. Write the regrouped line.”
- **Ask:** “Why is $(36 + 64) \times 7$ the same as $36 \times 7 + 64 \times 7$?” Draw 7 rows of dots split into a block of 36 and a block of 64. Expected: “Seven rows of 36 and seven rows of 64 make seven rows of 100.”
- **Decision point:** if a student cannot say why, stay on the array; do not move to Q5-6 on faith.
- **Check move:** “Estimate each product. Does 700 make sense?” (about $250 + 450$).
- **Name the trigger** and point to the strategy box: two products sharing a number → factor it out. Hand the pen to the student for Q6.

Worked Example 2 — Two methods, one answer: which is faster?

- **Split the room** (or, one-to-one, time the student twice). Half use Method A (calculate both sums), half use Method B (pairs). **Say:** “Go — hands up when done.” Record both times on the board.
- **Ask Method A:** “How did you get 650 for the evens?” Listen for $2 \times (1 + \dots + 25)$. If a student added term by term, note the time and do not correct yet.
- **Ask Method B:** “How many pairs? How do you know it is 25, not 24 or 50?” **Pause.** This is the counting decision; a student who says 50 has counted terms, not pairs.
- **Compare:** “Both answers are 25. Which method would you use at 30 seconds a question?” Expected secure answer: “Pairs — one count, no big additions.”
- **Transfer question:** “What if the evens went to 52 and the odds to 51?” (26 pairs → 26). Name the lesson: when two lists line up term by term, subtract in pairs.

6. Misconceptions (probe first, then correct)

Misconception	Probing question (student must say the reasoning aloud)	Correction
A longer decimal is a larger decimal.	Which is greater, 0.35 or 0.305? Convince me.	Pad to the same number of places (0.350 vs 0.305) and compare as whole numbers of thousandths.
Place names are counted from the left.	Point to the ten-thousands digit in 7 350 912. How did you find it?	Always count from the ones digit; group digits in threes from the right.
Subtracting a negative makes the answer smaller.	What is $9 - (-14)$? Show me on the number line.	Subtraction is the distance/difference: from -14 up to 9 is $14 + 9 = 23$. Rewrite $-(-a)$ as $+a$.
Farthest from 0 means the smallest number.	Which is smaller, -8.2 or 8.1 ? Which is farther from 0?	Distance from 0 ignores sign; ‘smallest’ uses sign. Say both questions aloud before answering.
Time is a decimal: 60 seconds behave like 100.	Write $48\text{ s} + 25\text{ s}$ in minutes and seconds.	Trade 60, not 100: $73\text{ s} = 1\text{ min } 13\text{ s}$. Circle the unit at each step.

Misconception	Probing question (student must say the reasoning aloud)	Correction
Calculate first, look later.	Before you multiply: what do 19×46 and 81×46 share?	Enforce a 10-second scan for a repeated factor, a round pair, or 25/125 partners before any algorithm.
$10^n - k$ has n nines.	Write out $10^4 - 47$. How many 9s? How many digits?	Build the small-case table; the last two digits come from $100 - 47$, so there are $n - 2$ nines.
A leftover part counts as a piece (or as a whole block when it should not).	From 100 cm, how many 30 cm pieces? From 88 min, how many 15-min blocks are paid?	Pieces round down (a leftover is waste); 'part of a block' charges round up. Read which one the question means.

7. Hint ladders for challenge items

Give one level at a time. The student writes the level used (H1, H2, H3) beside the item; log it in the error log so progress shows as fewer hints, not only more correct answers.

Item	H1 — reread, name a constraint	H2 — a representation, no calculation	H3 — one intermediate relationship
Sheet Q15	Reread: how many digits does 10^{30} have? How many does $10^{30} - 47$ have?	Make a table of $10^n - 47$ for $n = 2, 3, 4, 5$. Don't calculate the big one yet.	$10^n - 47$ is $(n - 2)$ nines followed by 53.
Sheet Q16	Reread: which pairs of factors make 10?	Rewrite 5^{20} as $5^2 \times 5^{18}$ and pair each 5 with a 2.	The product is 25×10^{18} .
Homework Q10	Reread: N is $10^{50} - 1$. What does that make $7 \times N$?	Make a table: 7×9 , 7×99 , 7×999 . Look at the digits, not the size.	$7 \times (10^n - 1) = 7 \times 10^n - 7$, which is 6, then $(n - 1)$ nines, then 3.

8. Differentiation

Struggling. Keep the reasoning, shrink the numbers: do Q3 with $-3^\circ\text{C} \rightarrow 4^\circ\text{C}$ on the printed number line, Q6 as $2 \times 5 + 8 \times 5$ with counters, and Q15 as $10^2 - 47$ and $10^3 - 47$ written out in full. Then return to the original numbers the same session. Use the place-value chart for every place-value item; use the number line for every integer item.

On-level. Complete Q1-14 at the 30-second target, then Q15 with the small-case table. Log every error with a code in the error log.

Advanced. For every independent item give a second method in the margin. Q15-16: write a one-line completeness argument (why the pattern holds for every n). Part C extension: find the digit sum of $10^{30} - 2\,026$, and of $13 \times (10^{50} - 1)$.

9. Per-item teacher notes

In-class sheet

Q	Answer	Concept tested	Likely error	Faster contest method
1	5	Place value: naming a place	3 (one place too far: hundred-thousands) or 9 (counted five places from the left)	Group digits in threes from the right; ten-thousands is the middle digit of the thousands group.
2	-8.2	Absolute value as distance from 0	-0.85 or 8.1 — chose the smallest-looking or the largest positive number	Compare only the sizes 8.2, 7.9, 0.85, 8.1, 8.02; the sign never matters.
3	17	Integer difference across zero	5 — computed $11 - 6$ (dropped the sign)	Distance across 0 = $11 + 6$.
4	8 min 13 s	Time arithmetic with regrouping	7 min 73 s or 8 min 73 s — no regrouping, or regrouped by 100	Round up: $3:48 + 4:25 = (4:00 - 12\text{ s}) + 4:25 = 8:25 - 12\text{ s} = 8:13$.
5	13 000	Structural rearrangement (convenient products)	$1\,625 \times 8$ done by hand and slipped (e.g. 12 000)	$125 \times 8 = 1\,000$, so the answer is $13 \times 1\,000$.
6	4 600	Factoring a common multiplier out of a sum	4 646 or an arithmetic slip from computing $874 + 3\,726$ by hand	$(19 + 81) \times 46 = 100 \times 46$.
7	(A) 0.305	Ordering decimals; middle value	(B) 0.35 — believes more digits means larger ($0.305 > 0.35$)	Pad to three places: 305, 350, 300, 053, 530 → middle 305.
8	(D) 28	Digit sums	(A) 19 — dropped the final 9; (B) 24 — dropped the leading 4	Skip the zeros and pair: $(4 + 8) + (7 + 9) = 12 + 16 = 28$.
9	(C) \$28.20	Decimal money $\times$ count	(B) \$25.85 — $10 \times 2.35 + 2.35$ (added one muffin, not two)	$12 \times 2.35 = 12 \times 2 + 12 \times 0.35 = 24 + 4.20$.
10	(B) 12	Division and comparison of counts	(A) 8 — subtracted the lengths $20 - 12$	$360 \div 12 = 30$ and $360 \div 20 = 18$; $30 - 18 = 12$.
11	(B) 40 350	Place value with powers of ten	(C) 43 050 — shifted the 3 one place left	Write a place-value row: 4 _ 3 5 _ → 40 350.
12	(C) 4	Integer addition and subtraction	(A) -10 — treated $-(-7)$ as -7	Subtracting -7 is adding 7: $12 + 7 - 15 = 4$.
13	(D) 4	Complement to a known total	(C) 3 — gave the top face ($7 - 4$) instead of the bottom face	Hidden = total - visible = $21 - 17$.
14	(E) 10:25 p.m.	Time arithmetic with regrouping	(A) 9:25 p.m. — did not carry the extra hour from 85 minutes	$7:48 + 12\text{ min} = 8:00$; $2\text{ h } 37\text{ min} - 12\text{ min} = 2\text{ h } 25\text{ min}$; $8:00 + 2:25 = 10:25$.

Q	Answer	Concept tested	Likely error	Faster contest method
15	(C) 260	Place value in $10^n - k$; digit sum of the result	(E) 270 — assumed 30 nines; (A) 252 — 28 nines but dropped the final 53	$10^{30} - 47 = (10^{30} - 1) - 46 = 28 \text{ nines followed by } 53 \rightarrow 28 \times 9 + 5 + 3.$
16	(B) 20	Pairing $5 \times 2 = 10$ (structural rearrangement with powers)	(D) 38 — added the exponents; (C) 21 — used 10^{20} instead of 25×10^{18}	$5^{20} \times 2^{18} = 5^2 \times (5 \times 2)^{18} = 25 \times 10^{18} \rightarrow 2 \text{ digits} + 18 \text{ zeros}.$
17	17 000	Convenient products	$4\,250 \times 4$ slipped (e.g. 16 000)	$250 \times 4 = 1\,000.$
18	4 000	Compensation around a round number	An addition slip from column-adding four 4-digit numbers	Each number is $1\,000 \pm \text{a little}$: $(-1) + (-2) + 2 + 1 = 0$, so $4 \times 1\,000.$

Homework

Q	Answer	Concept tested	Likely error	Faster contest method
1	(B) 80	<i>Retrieval.</i> Contest scoring geometry (Part A/B values)	(A) 75 — scored all 15 correct answers at 5 points	Part A is worth 50 in total; add $5 \times 6 = 30.$
2	(D) 400	<i>Retrieval.</i> Small cases reveal a pattern (sums of odd numbers are squares)	(E) 420 — summed $2 + 4 + \dots + 40$ (the evens); (A) 210 — summed $1 + 2 + \dots + 20$	Pair first and last: 20 terms make 10 pairs of 40 $\rightarrow 400.$
3	(B) 16	<i>Retrieval.</i> Working backward with inverse operations	(A) 11.5 — undid the steps in the wrong order: $(55 - 9) \div 4$	Undo in reverse: $55 + 9 = 64$, $64 \div 4 = 16$. Check: $16 \times 4 - 9 = 55.$
4	(E) 0.07	<i>Current topic.</i> Ordering decimals	(D) 0.0707 — more zeros and digits looked smaller	Pad to four places: 7000, 0700, 7070, 0770, 0707.
5	(A) 4	<i>Current topic.</i> Division and comparison of counts (with units)	(B) 10 — subtracted the lengths $40 - 30$	$480 \div 30 = 16$ and $480 \div 40 = 12.$
6	(C) \$10.50	<i>Current topic.</i> Time difference with regrouping; money $\times$ count; ceiling of a division	(A) \$8.75 — ignored the part block (5 blocks for 88 min)	$9:52 \rightarrow 11:20$ is 1 h 28 min = 88 min; 90 min is 6 blocks; $6 \times 1.75 = 10.50.$
7	(C) 8°C	<i>Current topic.</i> Integer change on a number line	(D) 20°C — started from +6 instead of -6	Net change $-9 + 23 = +14$; $-6 + 14 = 8.$
8	(B) 7:18 a.m.	<i>Interleaved.</i> Work backward through time with regrouping	(C) 7:22 a.m. — forgot the 4 minutes early	Total 59 min = 1 h - 1 min: $8:17 - 1 \text{ h} = 7:17$, + 1 min = 7:18.

Q	Answer	Concept tested	Likely error	Faster contest method
9	(E) 95	<i>Interleaved.</i> Sum of consecutive numbers = count $\times$ middle value; eliminate options	(B) 63 or (D) 81 — tried a few sums, found none, and guessed	Five consecutive numbers sum to $5 \times$ middle, so the sum is a multiple of 5.
10	(C) 450	<i>Challenge.</i> $10^n - 1$ structure; digit sum	(A) 350 — assumed the answer is 7×50	$7 \times N = 7 \times 10^{50} - 7 = 6$, then forty-nine 9s, then $3 \rightarrow 6 + 49 \times 9 + 3$.

Drill book: answers are on its tear-off key. Drill items carry no teacher notes by design — they are fluency repetitions, not problems.

10. Exit check and parent update

Lesson A exit: (1) $25 \times 37 \times 4 = 3\,700$. (2) -6.5 is farther from 0 than 6.05. **Lesson B exit:** Sheet Q17–18 (17 000; 4 000 with a compensation explanation). Pass mark for moving on: 2/2 on both exits and 90% on Drill Set 2 inside 6 minutes; otherwise repeat Drill Sets 1–2 in the extension session before Set 3.

Parent update line: “[Child] worked on integers, place value and fast mental arithmetic for contest Part A; strongest at ____; we will practise ____ next, with timed drill sets at home (10 minutes a day).”